

Closing the Confidence Gap

A Product Improvement Initiative for Pearson MyLab | Moukthika Vasa | MBA '26, Darla Moore School of Business, USC

D — DEFINE

Problem Statement

Pearson's MyLab captures confidence through its answer selection mechanism. When a student selects a specific answer but gets it wrong, MyLab flags it as "YOU WERE SURE AND INCORRECT" — yet responds with static, generic remediation. The system knows WHAT went wrong but cannot diagnose WHY. Instructors receive no proactive alert.

Project Scope

- Platform: Pearson MyLab — all disciplines
- Users: College students & instructors (Higher Ed)
- Focus: Confidence-based gap detection & instructor alerting

Goal Statement

Design and deploy a Misconception Intelligence Program within MyLab that moves beyond algorithmic repetition — providing root-cause-targeted interventions for students and proactive confidence pattern alerts for instructors.

M — MEASURE

How MyLab Measures Confidence (3-Step Process)

Step 1 — Confidence via Answer Selection:

Confidence is embedded in the answer choices themselves. Selecting a specific answer (e.g. 101.6) signals confidence. Selecting the option labelled I DON'T KNOW YET signals uncertainty. No separate rating step required.

Step 2 — Calibration Check:

System compares stated confidence vs. actual accuracy. Sure + Wrong → flagged as "YOU WERE SURE AND INCORRECT." Mastery score penalized heavily.

Step 3 — Adaptive Remediation:

Student directed to review activities or cooldown. Same pre-written content regardless of specific wrong answer.

Current State Gap:

No differentiated response for all 4 confidence-accuracy scenarios. Instructors must manually review individual results — no aggregated at-risk dashboard.

A — ANALYZE

4 Confidence-Accuracy Scenarios

#	Confidence	Answer	Meaning
1	High	Wrong	Misconception — flawed mental model
2	High	Correct	True mastery — no intervention needed
3	Low	Wrong	Knowledge gap — standard remediation
4	Low	Correct	Test anxiety — needs confidence coaching

3 Root Causes

RC1: No misconception root-cause analysis — MyLab knows WHICH wrong answer but not the logic path that led there.

RC2: Static help system — "Help Me Solve This" shows generic correct steps, not feedback tied to the student's specific wrong answer.

RC3: No instructor proactive alerting — no aggregated confidence-accuracy dashboard; instructors must manually review results.

I — IMPROVE — Proposed Solution: MyLab Misconception Intelligence Program

1 Workstream 1 — Root Cause Feedback Engine

When a student is flagged Sure + Wrong, instead of generic correct steps, the system provides feedback tied to their specific wrong answer: "You answered 14 — this suggests you added instead of multiplied. Here is why that approach doesn't apply."

- Requires wrong-answer taxonomy mapping per question type
- Initial implementation: structured taxonomy → Future: LLM-based diagnostic analysis

2 Workstream 2 — Instructor Confidence Intelligence Dashboard

New instructor-facing dashboard aggregates confidence-accuracy data and surfaces actionable insights:

- Weekly digest: students with repeated Sure+Wrong patterns in same concept
- Class-level heatmap: concepts with highest overconfidence rate
- Recommended intervention type per student (which scenario applies)
- Instructor responds directly — no manual result review required

3 Workstream 3 — Confidence Calibration Track (Scenario 4)

Students who consistently show Low Confidence + Correct are anxiety-gap students, not knowledge-gap students. Current MyLab treats them identically.

- Separate "Confidence Calibration" module for Scenario 4
- Exercises reinforcing correct answers and building self-trust
- Instructor alert: motivational intervention signal, not academic

12-Week Program Roadmap

DISCOVERY	DESIGN	BUILD & TEST	DEPLOY
Weeks 1–2 Stakeholder interviews. Wrong-answer taxonomy for pilot subject.	Weeks 3–5 Root Cause Feedback Engine logic. Instructor Dashboard wireframes.	Weeks 6–9 Engineering build. UAT with pilot instructors and students.	Weeks 10–12 Phased rollout. Instructor training. KPI monitoring begins.

C — CONTROL

Success Metrics

Metric	Baseline	Target
Repeated Sure+Wrong same concept	No tracking	↓ 25% in 4 weeks
Instructor time to find at-risk students	Days–weeks (manual)	< 48 hrs (automated)
Concept mastery post-intervention	No measurement	↑ 20% in 2 weeks
Scenario 4 confidence score	Not tracked	↑ 15% self-reported

Sustainability Plan

- Quarterly review of confidence-accuracy patterns at discipline level
- Continuous wrong-answer taxonomy expansion as new questions added
- Annual instructor survey on dashboard utility and adoption

Conclusion

Future: LLM-based root cause analysis replacing structured taxonomy
MyLab collects confidence data but hasn't closed the loop to meaningful intervention. This program moves from identifying confident mistakes to understanding & resolving them.